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Performance optimization of active RIS-assisted UAV communications

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ABSTRACT

Unmanned aerial vehicles (UAVs) are considered a promising technology for the future sixth-generation (6G) systems because of their advantages, such as extended coverage, energy efficiency, concentrated signal, adaptability, and capacity to offer ground terminal computing services. Utilizing UAVs can greatly minimize compute task latency and improve system performance. Furthermore, Reconfigurable Intelligent Surfaces (RISs) are an emerging technology that has the potential to improve wireless networks' propagation environments. UAV-assisted RIS communications are considered a potential solution for future generations of networks. We consider a downlink communication scenario where an UAV operating as a base station is assisted by an active RIS. By simultaneously adjusting the UAV beamforming and RIS coefficients, we hope to maximize the sum rate of users. We propose an iterative algorithm for solving this problem. Then, we analyse the system performance in terms of total rate, energy efficiency, and robustness to channel estimate errors, while explicitly accounting for the power consumption of active RIS components. A comparison with a standard passive RIS-assisted UAV network is also conducted. Numerical results confirm that the active RIS compensates for the considerable path loss associated with UAV-to-ground channels, hence significantly increasing the sum rate. Additionally, the suggested system provides improved energy efficiency in the low-to-moderate power regime, owing to the superior spectrum efficiency given by signal amplification

1. Introduction

Reconfigurable Intelligent Surface (RIS) has just become a prospective solution to meet the growing needs of communication and mobile devices. RIS uses a smart radio environment to improve networks' spectral and energy efficiency. These surfaces have several reflecting components that can change their reflecting characteristics to improve network efficiency and provide greater control over signal propagation [1–4]. RISs can reduce signal degradation and boost coverage by optimizing signal reflections and signal direction management. This improves signal quality and increases data rates for receivers. Additionally, an RIS may assist in optimizing resource allocation, interference management, and enhancing overall network performance [5–8]. The idea behind RIS is to use reflecting surfaces that may be adjusted in terms of phase or amplitude of incident or reflected signals. RISs can successfully counteract interference issues and fading environment impairments by smartly modifying the features of these surfaces [9,10]. According to developments in metasurfaces and micro-electromechanical systems (MEMS), controlled phase shifters may now be used to alter RIS components in real-time. Its dynamic reconfiguration enables flexibility in response to different

environmental factors and communication settings. For instance, many publications used the single passive RIS to enhance network functionality [11,12].

Although using single and multiple passive RISs to increase network performance has many benefits, one particular drawback for passive RIS-assisted systems is double-fading attenuation. This is because reflected signals have to go through a cascaded channel made up of transmitter-user and RIS-user links. This restriction greatly reduces the receiver's ability to receive signals and the effective coverage of each RIS [13]. Then active RIS, has been introduced and garnered a lot of interest. By regulating the phase and amplitude of the reflected signal, active RIS can expand the system's capacity. Many research has treated active RIS in various fields. Active RIS is considerably cheaper compared to other repeaters (such as amplify-and-forward (AF) relay and decode-and-forward (DF) relay), despite including a low-power amplifier at a little greater cost than passive RIS. With this focus, the author of [14] demonstrated that ARIS can convert the double-fading path loss of PRIS into the additive version. Afterwards, the sub-connected structure of ARIS was introduced to control the phase shift of each element separately [15]. The use of ARIS in wireless communications was described

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in Long et al. [16], where an increase in active elements increased the ARIS-assisted links. The authors in [17] present the use of active RIS in non-orthogonal multiple access (NOMA) systems where hardware impairments are incorporated. They get the precise and asymptotic outage probability equations for ARIS-NOMA-HIS networks across cascade Nakagami-m fading channels. They further get diversity orders of ARIS-NOMA-HIS networks, which are associated with both channel ordering and the quantity of reflecting components, by applying theoretical analysis. They calculate the outage probability for ARIS-OMA-HIS networks as a reference point. The authors in Yue et al. [18], presents the active simultaneously transmitting and reflecting surface (ASTARS) to support NOMA communications, in which the service area's paring users are evenly distributed. In order to align the phases of cascaded channels at pairing users, they construct the separate reflection/transmission phase-shift controllers of ASTARS. For ASTARS-NOMA networks with perfect/imperfect successive interference cancellation (pSIC/ipSIC), they obtain new approximate and asymptotic estimates of the outage probability and ergodic data rate.

Moreover, taking into account the remarkable maneuverability and simple installation of unmanned aerial vehicles (UAVs) [19,20], the RIS installed with the UAV is presented as a supplementary level of freedom (DoF) for enhancing network efficiency [21,22]. UAVs have emerged as a promising means of enhancing air-ground communication quality due to their low cost, high flexibility, and potential for line-of-propagation (LoP) links. UAVs can generally be used for data collection, tracking, and other functions in addition to serving as base stations, relays, and transmitters of signals. RIS-assisted UAV systems have been greatly studied due to their advantages. In Hua et al. [23], the authors examined a symbiotic UAV-aided RIS radio network and presented a relaxation-based approach to reduce the weighted aggregate bit error rate of all RISs. Two strategies were put out to jointly derive a UAV trajectory and use active and passive beamforming to enhance the spectrum and energy efficacy of an RIS-aided UAV network. The suggested methods produced greater performance than the baselines. In physical layer security (PLS), UAV-assisted RIS networks play an important role. UAVs have a higher chance of creating robust line-of-sight (LoS) channels to the ground terminal. The private information is vulnerable to eavesdropping under LoS link settings because the wiretap channels that are connected to it are usually LoS-dominant. Due to its notable performance benefits, the RIS has been extensively studied in the latest research and has emerged as the best solution for adopting and improving PLS. The RIS may reduce the received signal strength for eavesdroppers by strengthening signals at the intended receiver and decreasing or removing signals in non-targeted destinations by varying the phases and amplitudes of the signals.

Recently, a lot of research has been done on RIS-assisted UAV communications [24,25]. In particular, Cao et al. [26] created an adaptive RIS-assisted transmission protocol to use machine learning and numerical methods to optimize the system throughput. Although the previously listed studies have established a strong basis for UAV-aided RIS communications, there is still much to learn about implementing RIS in UAV-enabled systems, particularly in scenarios involving many users. The joint UAV precoding and RIS reflection coefficient optimization problem has been the subject of various research efforts [27], but the system concepts that have been considered are only suitable for single-user scenarios; multiple access strategies are not considered and active RIS is not treated.

A static UAV makes channel modeling and RIS phase-shift optimization easier. Since the UAV-RIS-user geometry is fixed, static UAVs are more suited for long-term deployments such as disaster recovery and rural coverage since they use less power than mobile ones. Static UAV combined with RIS presents a strong argument for long-term wireless coverage, especially in situations where continuous connectivity is necessary. Three main benefits of this architecture are (i) a significant reduction in energy consumption due to the elimination of propulsion requirements, (ii) a simplified estimation of channel state through fixed

geometry, and (iii) a computationally efficient RIS configuration that does not require dynamic beam tracking. The static UAV-RIS paradigm is especially well-suited for three crucial application domains because of these features: large-scale Internet of Things (IoT) deployments that require stable channel conditions, emergency communication systems that need to be deployed quickly, and rural area networks with limited infrastructure. For long-term deployments, the stationary technique offers an alluring trade-off by lowering operational complexity while preserving coverage reliability when compared to mobility UAV-assisted systems.

The main distinctions between traditional terrestrial networks and active RIS-assisted UAV architecture systems are the altitude-enabled line-of-sight connectivity and adjustable coverage extension. While terrestrial networks are limited by fixed infrastructure and multipath-rich situations, a UAV positioned at a fixed altitude can provide stronger LoS communication- and modify the RIS to optimize reflections for ground users. The probability of LoS connectivity to both the RIS and ground users is significant while a UAV is stationary. Terrestrial base stations, on the other hand, are limited by structures, topography, and clutter, which frequently cause significant multipath and shadowing. The RIS can be actively set up to take use of the UAV's advantageous point in UAV-assisted scenarios, directing reflections toward targeted ground terminals. In contrast to terrestrial RIS deployments, which are constrained by fixed angles of incidence and reflection, this allows for more effective beamforming. Moreover, to provide coverage in places without or with blocked terrestrial infrastructure, a static UAV can be positioned strategically. In contrast to terrestrial RIS channels, which are dominated by rich multipath, UAV-RIS channels show lower scattering and altitude-dependent path loss. The development of active RIS control algorithms is directly impacted by this distinction.

Although passive RIS can increase coverage by intelligently reflecting signals, it merely redistributes existing energy and can't compensate for significant path loss. Active RIS, on the other hand, has amplification characteristics that allow the system to both enhance and alter the signal. This is especially important in UAV-assisted settings, where altitude and distance frequently result in considerable free-space path loss in the air-to-ground channel. By increasing the received signal power, Active RIS offers a way to get over these restrictions and guarantee more dependable connectivity and wider coverage. When UAV-based links experience severe attenuation, active RIS is not only an improvement but an essential remedy for reducing path loss in UAV communications. Accordingly, to provide enhanced aerial communication capabilities, we present an architecture in which a UAV serves as a base station with assistance from an active RIS. The primary characteristic of active RISs is their capacity to reflect signals with amplification, which may be achieved by incorporating reflection-type amplifiers. This sets them apart from traditional RISs, which simply reflect signals without amplification.

The main contributions in this paper can be summarized as :

- we propose a scenario deploying a UAV serving as a base station and assisted active RIS, and we suggest jointly optimizing UAV beamforming and RIS precoding approach to maximize the network sum-rate under transmit power constraints.
- A comprehensive performance evaluation of UAV-assisted active RIS networks under realistic conditions is treated by rigorously investigating sum-rate (SR), energy efficiency, and the effects of channel estimation errors.
- The proposed iterative algorithm relies on an alternating optimization approach and focuses on jointly optimizing the RIS precoding and UAV beamforming.
- Results prove that active RIS can significantly enhance the SR of users and resilience to imperfect channel knowledge, while also revealing significant performance on energy efficiency in UAV-assisted active RIS systems compared to their passive RIS counterparts.

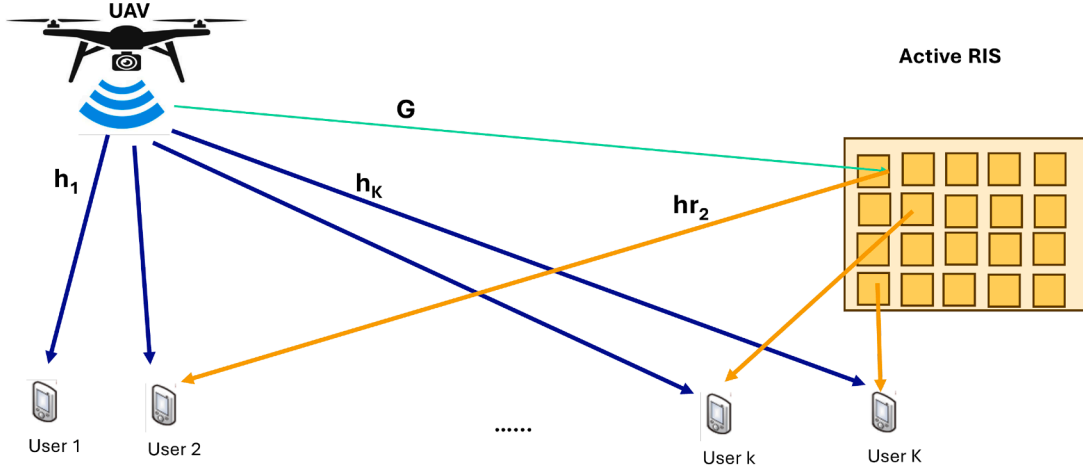


Fig. 1. Active RIS-assisted UAV communications..

2. System model and problem formulation

2.1. System model

We examine an UAV-assisted active RIS that is equipped with N_r antennas and serves K single-antenna users in downlink scenario. The active RIS is mounted at a specific altitude on the building front and supports air-ground communication while the UAV is positioned at a constant altitude of z_0 . Let the set of UEs be represented by $\zeta = 1, 2, \dots, K$. The transmitted symbol vector utilised for all users is represented as $\mathbf{s} := [s_1, \dots, s_K]^T \in \mathbb{C}^K$, where $\mathbb{E}[\mathbf{s}\mathbf{s}^H] = \mathbf{I}_K$. The received signal $y_k \in \mathbb{C}$ at user k is expressed as follows :

$$\mathbf{y}_k = \underbrace{(\mathbf{h}_k^H)}_{\text{UAV-user link}} + \underbrace{(\mathbf{h}_k^H \mathbf{\Psi} \mathbf{G})}_{\text{RIS-user link}} \sum_{j=1}^K \mathbf{w}_j s_j + \underbrace{(\mathbf{h}_k^H \mathbf{\Psi} \mathbf{v})}_{\text{Noise inserted by active RIS}} + \underbrace{n_k}_{\text{Noise inserted at user } k}, \quad (1)$$

We denote by $\mathbf{h}_k^H \in \mathbb{C}^{1 \times N_t}$, $\mathbf{G} \in \mathbb{C}^{N \times N_r}$ and $\mathbf{h}_k^H \in \mathbb{C}^{1 \times N}$, the channels between the UAV and UE k , between the UAV and the RIS, and that between the RIS and UE k , respectively. N and N_t represent the number of RIS elements and transmit antennas at the UAV, respectively. The UAV beamforming vector for each symbol s_k is represented as $\mathbf{w}_k \in \mathbb{C}^{M,1}$, while n_k represents the Additive White Gaussian Noise (AWGN) at UE k .

2.2. Problem formulation

We examine a UAV-assisted active RIS in a MU-MISO system. In this subsection, we determine the SR maximization problem. Then, we introduce a joint transmit beamforming and reflect precoding algorithm to solve the issue. Based on (1), the Signal-to-Interference-plus-Noise Ratio (SINR) at user k is represented as follows :

$$\text{SINR}_k = \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum_{j=1, j \neq k}^K |\mathbf{h}_k^H \mathbf{w}_j|^2 + \|\mathbf{h}_k^H \mathbf{\Psi}\|^2 \sigma_v^2 + \sigma^2}, \quad (2)$$

where $\bar{\mathbf{h}}_k = \mathbf{h}_k^H + \mathbf{h}_k^H \mathbf{\Psi} \mathbf{G} \in \mathbb{C}^{1 \times M}$ represents the equivalent channel between UAV and UE k , which involves both the direct and reflected channels. The UAV transmit power denoted P_{UAV} and the power of ac-

tive RIS P_{RIS} are expressed as:

$$P_{\text{UAV}} = \mathbb{E} \left\{ \left\| \sum_{k=1}^K \mathbf{w}_k s_k \right\|^2 \right\} = \underbrace{\sum_{k=1}^K \|\mathbf{w}_k\|^2}_{\text{Desired signal power}}, \quad (3a)$$

$$P_{\text{RIS}} = \mathbb{E} \left\{ \left\| \mathbf{\Psi} \mathbf{G} \sum_{k=1}^K \mathbf{w}_k s_k + \mathbf{\Psi} \mathbf{v} \right\|^2 \right\} = \underbrace{\sum_{k=1}^L \|\mathbf{\Psi} \mathbf{G} \mathbf{w}_k\|^2}_{\text{Desired signal power}} + \underbrace{\|\mathbf{\Psi}\|_F^2 \sigma_v^2}_{\text{Amplified noise power}}. \quad (3b)$$

The reflective power of the active RIS P_{RIS} should compensate for the increased power consumption resulting from the noise amplification, in contrast to the UAV transmit power P_{UAV} , which only contains the required signal power.

Thus, considering the power limitations at the UAV and the active RIS, the initial SR maximization problem can be stated as follows:

$$P_1 : \max_{\mathbf{w}, \mathbf{\Psi}} \text{SR}(\mathbf{w}, \mathbf{\Psi}) = \sum_{k=1}^K \log_2 (1 + \text{SINR}_k), \quad (4a)$$

$$\text{s.t.} : \sum_{k=1}^K \|\mathbf{w}_k\|^2 \leq P_{\text{UAV}}^{\max}, \quad (4b)$$

$$: \sum_{k=1}^K \|\mathbf{\Psi} \mathbf{G} \mathbf{w}_k\|^2 + \|\mathbf{\Psi}\|_F^2 \sigma_v^2 \leq P_{\text{RIS}}^{\max}, \quad (4c)$$

$\mathbf{w} := [\mathbf{w}_1^T, \dots, \mathbf{w}_K^T]^T$ is the beamforming vector of UAV. The power constraints at the UAV and active RIS are (4b) and (4c), respectively. Since problem P_1 is non-convex and presents correlated variables, it is difficult to optimize $\mathbf{w}$ and $\mathbf{\Psi}$ jointly.

We determine a combined beamforming and precoding system based on fractional programming (FP) and alternating optimization (AO) in order to effectively handle this issue.

2.3. Proposed algorithm

To tackle the problem effectively, we first reformulate it. The RIS precoding matrix is represented as $\mathbf{\Psi}$ and the UAV beamforming vector expressed as $\mathbf{w}$. To handle the non-convex function in (4), we use the FP techniques suggested in [28] to decouple the variables in problem P_1 . Auxiliary variables are introduced: $\mathbf{l} := [\mathbf{l}_1, \dots, \mathbf{l}_K] \in \mathbb{R}_+^K$ and

$\mathbf{t} := [\mathbf{t}_1, \dots, \mathbf{t}_K] \in \mathbb{R}^K$, problem P_1 in (11) can be expressed such :

$$P_2 : \max_{\mathbf{w}, \Psi, \mathbf{l}, \mathbf{t}} SR(\mathbf{w}, \Psi, \mathbf{l}, \mathbf{t}) = \sum_{k=1}^K \ln(1 + l_k) - \sum_{k=1}^K l_k + \sum_{k=1}^L g(\mathbf{w}, \Psi, l_k, t_k),$$

$$\text{s.t. } C_1 : \|\mathbf{w}\|^2 \leq P_{\text{UAV}}^{\max},$$

$$C_2 : \sum_{k=1}^K \|\Psi \mathbf{G} \mathbf{w}_k\|^2 + \|\Psi\|_F^2 \sigma_v^2 \leq P_{\text{RIS}}^{\max}, \quad (5)$$

while $g(\mathbf{w}, \Psi, \mathbf{l}_k, \mathbf{t}_k)$ is determined as :

$$g(\mathbf{w}, \Psi, \mathbf{l}_k, \mathbf{t}_k) = 2\sqrt{(1 + l_k)} \Re\{\mathbf{t}_k^* \bar{\mathbf{h}}_k^H \mathbf{w}_k\} - |t_k|^2 \left(\sum_{j=1}^K |\bar{\mathbf{h}}_k^H \mathbf{w}_j|^2 + \|\mathbf{h}_k^H \Psi\|^2 \sigma_v^2 + \sigma^2 \right). \quad (6)$$

As demonstrated in [28] that the FP approaches converged strongly. A locally optimal solution to P_2 can therefore be produced by alternately improving the auxiliary variables $\mathbf{l}$ and $\mathbf{t}$, the RIS precoding matrix Ψ , and the beamforming vector $\mathbf{w}$ in each iteration step until $\mathbf{SR}$ converges, provided that all of the updates in each iteration step are optimal. The proposed beamforming and precoding algorithm is summarized in Table 1. The optimal solutions for $\mathbf{w}$, Φ , $\mathbf{l}$, and $\mathbf{t}$ are determined, respectively.

2.3.1. Optimizing auxiliary variables

After fixing the others variables ($\mathbf{w}, \Psi, \mathbf{t}$), the optimum solution l_{opt} can be found by solving : $\frac{\partial SR}{\partial l} = 0$ which is expressed as :

$$l_{\text{opt}}^k = \frac{\beta_k^2 + \beta_k \sqrt{\beta_k^2 + 4}}{2}, \quad \forall k \in \{1, \dots, K\}, \quad (7)$$

where $\beta_k = \Re\{\mathbf{t}_k^* \bar{\mathbf{h}}_k^H \mathbf{w}_k\}$.

Similarly, after fixing $(\mathbf{w}, \Psi, \mathbf{l})$, the optimal solution of $\mathbf{t}$ is determined by

$$t_k^{\text{opt}} = \frac{\sqrt{(1 + l_k)} \bar{\mathbf{h}}_k^H \mathbf{w}_k}{\sum_{j=1}^K |\bar{\mathbf{h}}_k^H \mathbf{w}_j|^2 + \|\mathbf{f}_k^H \Psi\|^2 \sigma_v^2 + \sigma^2}, \quad \forall k \in \{1, \dots, K\}. \quad (8)$$

2.3.2. Optimizing the UAV beamforming

Then we optimize $\mathbf{w}$ after fixing $(\Psi, \mathbf{l}, \mathbf{t})$.

Problem P_2 in (12) can therefore be rewritten as follows:

$$P_3 : \max_{\mathbf{w}} \Re\{2\mathbf{b}^H \mathbf{w}\} - \mathbf{w}^H \mathbf{S} \mathbf{w},$$

$$\text{s.t. } C_1 : \|\mathbf{w}\|^2 \leq P_{\text{UAV}}^{\max},$$

$$C_2 : \mathbf{w}^H \mathbf{D} \mathbf{w} \leq P_{\text{m}}^{\max}. \quad (9)$$

where :

$$\mathbf{b}_k^H = \sqrt{(1 + l_k)} t_k^* \bar{\mathbf{h}}_k^H, \mathbf{b} = [\mathbf{b}_1^T, \mathbf{b}_2^T, \dots, \mathbf{b}_N^T]^T, \quad (9a)$$

$$\mathbf{S} = \mathbf{I}_K \otimes \sum_{k=1}^K |t_k|^2 \bar{\mathbf{h}}_k \bar{\mathbf{h}}_k^H, \mathbf{D} = \mathbf{I}_K \otimes (\mathbf{G}^H \Psi^H \Psi \mathbf{G}), \quad (9b)$$

$$P_{\text{m}}^{\max} = P_{\text{RIS}}^{\max} - \|\Psi\|_F^2 \sigma_v^2. \quad (9c)$$

Considering that P_3 in (9), is a Quadratic Constraint Quadratic Programming (QCQP) problem, the optimal solution $\mathbf{w}_{\text{opt}}$ to P_3 in (9) can be reformulated by exploiting the Lagrange multiplier method as:

$$\mathbf{w}^{\text{opt}} = (\mathbf{S} + \lambda_1 \mathbf{I}_{MK} + \lambda_2 \mathbf{D})^{-1} \mathbf{b}, \quad (10)$$

λ_1 and λ_2 represent the Lagrange multipliers, determined considering the complementary slackness requirements for power constraints C1 and C2. $\lambda_{\text{opt}1}$ and $\lambda_{\text{opt}2}$ can be found :

$$P_4 : \max_{\Psi} \Re\{2\mathbf{v}^H \mathbf{u}\} - \mathbf{v}^H \alpha \mathbf{v},$$

$$\text{s.t. } C_2 : \mathbf{v}^H \mathbf{v} \leq P_{\text{RIS}}^{\max}, \quad (11)$$

wherein

$$\mathbf{v} = \sum_{k=1}^K \sqrt{(1 + l_k)} \text{diag}(t_k^* \bar{\mathbf{h}}_k^H) \mathbf{G} \mathbf{w}_k - \sum_{k=1}^K |t_k|^2 \text{diag}(\mathbf{h}_k^H) \mathbf{G} \sum_{j=1}^K \mathbf{w}_j \mathbf{w}_j^H \mathbf{h}_k, \quad (11a)$$

$$\alpha = \sum_{k=1}^K |t_k|^2 \text{diag}(\mathbf{h}_k^H) \text{diag}(\mathbf{h}_k) \sigma_v^2 + \sum_{k=1}^K |t_k|^2 \sum_{j=1}^K \text{diag}(\mathbf{h}_k^H) \mathbf{G} \mathbf{w}_j \mathbf{w}_j^H \mathbf{G}^H \text{diag}(\mathbf{h}_k), \quad (11b)$$

$$\mathbf{v} = \sum_{k=1}^K \text{diag}(\mathbf{G} \mathbf{w}_k) (\text{diag}(\mathbf{G} \mathbf{w}_k))^H + \sigma_v^2 \mathbf{I}_N. \quad (11c)$$

Considering that (11) represents a QCQP problem, then to find $\mathbf{v}^{\text{opt}}$, which is the optimal solution, the Lagrange multiplier is used as follows:

$$\mathbf{v}^{\text{opt}} = (\alpha + \mu \mathbf{I})^{-1} \mathbf{v}, \quad (12)$$

We define μ as the Lagrange multiplier, which must be selected to satisfy the power constraint C2. To determine the optimal solution μ^{opt} , we use a binary search.

Algorithm 1 Iterative UAV-active RIS algorithm.

- 1: **Input:** Channels: G, h_k and $h_{r,k}, k \in \{1, \dots, K\}$ maximum transmission power of UAV $P_{\text{UAV}}^{\max}$
- 2: **Initialization:**
- 3: set $i = 0$,
- 4: RIS precoding vector Ψ^0 ,
- 5: UAV beamforming vector $\mathbf{w}^0$,
- 6: auxiliary variables l^0, t^0 ,
- 7:
- 8: **repeat:** until convergence of SR;
- 9: Update intermediate variable l by (7);
- 10: Determine auxiliary variable t_k^{opt} via (8);
- 11: Calculate $\mathbf{w}_{\text{opt}}$ by solving (10);
- 12: Update Ψ based on (11);
- 13: Determine SR expressed as: $SR(\mathbf{w}, \Psi) = \sum_{k=1}^K \log_2(1 + SINR_k)$;
- 14: update $i = i + 1$
- 15: **end**
- 16: **return :** $\mathbf{w}^{\text{opt}}, \Psi^{\text{opt}}, SR$.
- 17:

3. Computational complexity analysis

The proposed approach for resolving the maximum sum-rate issue in an ARIS-assisted UAV system is based on an iterative optimization strategy. This framework switches between optimizing the ARIS precoder and the BS transmit beamformer, two related sub-problems. $N_r K$ complex variables are used in the first subproblem. Power restrictions related to the SR are among the constraints in this sub-problem. Typically, matrix operations like inversion are required to solve this convex problem, resulting in a computational complexity of $O((N_r K)^3)$. The second subproblem is about complex variables. Similar to the beamforming subproblem, it has quadratic restrictions pertaining to the SR needs. The computational complexity of addressing this problem is roughly equal to $O(K^2 N N_r)$. Therefore, it is possible to express the algorithm's overall complexity per iteration as $O((K^2 N N_r) + ((N_r K)^3))$. The Overall complexity is: $O(I((K^2 N N_r) + ((N_r K)^3)))$ where I is the number of iteration required for convergence.

4. Simulation results

Numerical results are discussed in this part to verify [Algorithm 1](#) and the active RIS's efficiency [29,30]. It is assumed that the RIS is installed at a fixed altitude of $z_0 = 100\text{m}$ and the UAV operates at a constant altitude of $z_1 = 200\text{m}$. The number of UE is $K = 4$ while the users are uniformly and randomly separated in an area of radius 100m^2 around the RIS.

The direct UAV-UE link is modeled using Rayleigh fading, the UAV-RIS and RIS-UE channels employ Rician fading with Rician factors of 100 and 10, respectively. Our framework concentrates on the theoretical performance under these conventional fading models, while acknowledging that more realistic UAV channel modeling such as geometry based stochastic models or measurement driven models represents a significant avenue for future research. Big buildings often block the LoS links between the UAV and ground users, whereas the non-LoS scattering components remain significant, particularly in low-to-medium frequency bands and dense urban environments. The path loss for a given link distance d is expressed as PL.

$$PL = 41.2 + 28.7 \log d, \quad (13)$$

where d is the distance between two devices. The RIS-user and BS-RIS channels are generated by PL. For all channels concerned, following [31] we use the Ricean fading channel model to adjust for small-scale fading. Consequently, a randomly selected channel matrix H is defined by:

$$\mathbf{H} = \sqrt{PL} \left(\sqrt{\frac{\kappa}{\kappa+1}} \mathbf{H}_{LoS} + \sqrt{\frac{1}{\kappa+1}} \mathbf{H}_{NLoS} \right), \quad (14)$$

while PL represents the path loss of H ; κ describes the Rician factor; and H_{LoS} and H_{NLoS} represent the deterministic LoS and Rayleigh fading components. In particular, here we assume $\kappa = 1$.

The UAV and the active RIS are placed at (0, 200 m) and (10 m, 100 m), respectively. The locations of the users will be prescribed eventually. The number of UAV antennas and RIS elements are arranged as $N_t = 4$ and $N = 64$, respectively. The noise power is set as $\sigma^2 = \sigma_v^2 = -100\text{dBm}$. Let $P_{UAV}^{\max}$ define the maximum transmit power at the UAV and $P_{RIS}^{\max}$ represents the maximum reflect power of the active RIS. We restrict the total power consumption: $P_{\max} = P_{UAV}^{\max} + P_{RIS}^{\max}$ to 10dBm by setting $P_{UAV}^{\max} = 0.99 P_{\max}$ and $P_{RIS}^{\max} = 0.01 P_{\max}$ for the active RIS scenario, and $P_{UAV}^{\max} = 10\text{dBm}$ for the other benchmark scenarios. We examine the following scenarios for simulations:

UAV-Aided Active RIS: An UAV-aided active RIS scenario is examined, where the proposed [Algorithm 1](#) is utilized to jointly optimize the UAV beamforming and the precoding at the active RIS.

UAV aided-passive RIS: An UAV-aided passive RIS system, while the algorithm proposed in [32] is considered to jointly optimize the UAV beamforming and the precoding at the passive RIS.

UAV Without RIS: An UAV scenario without RIS, the WMMSE approach from [33] is considered to optimize the UAV beamforming.

4.1. Achieved sum rate versus the total power

[Fig. 2](#) examines the effect of increasing the transmit power on the sum rate for UAV-assisted active RIS and other benchmarks. The figure demonstrates that system performance improves as the transmit power increases. According to the results presented in [Fig. 2](#), UAV-aided active RIS is still able to outperform UAV-passive RIS in terms of SNR gain, despite the active components introducing extra thermal noise. This occurs because, although the injected noise terms are not coherently generated with the same phase at the user, the required signals reflected by various active RIS elements are added coherently. Furthermore, the RIS-user path loss has made these generated noises much smaller when they are received by the user. Furthermore, in contrast to the passive RIS-assisted system, which experiences multiplicative path loss of reflected links in all radiation power, the "multiplicative fading" issue can be effectively avoided and power degradation is significantly reduced

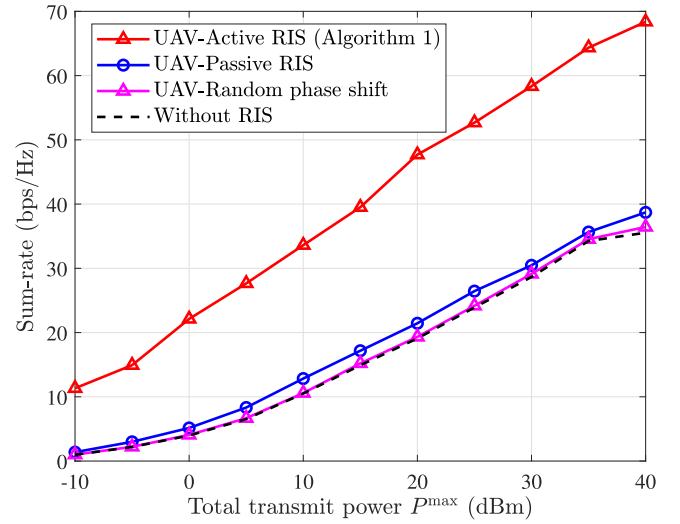


Fig. 2. Achieved sum rate versus the total power with $K = 4, N = 64, D = 50, 200$.

because the power radiated by active RISs only encounters the substantial fading of the RIS-user link.

4.2. Sum rate of the UEs in UAV-aided RIS system versus number of RIS elements

[Fig. 3](#) demonstrates the effect of increasing the number of RIS elements N in the sum rate for active RIS and other benchmarks. The figure demonstrates that system performance improves as the number of deployed reflecting elements augments, with the active RIS benefiting more from this improvement. The active RIS scenario achieves for example, 30bits/s/Hz when $N = 500$ while the passive RIS-based counterpart achieves only bits/s/Hz . This enhancement is assigned to the active RIS's ability to increase the reflected signals, compensating for substantial route loss and increasing signal strength. Moreover, as N increases, the performance improvement of the UAV-assisted active RIS scheme gradually stagnates.

4.3. Sum rate of the UEs versus the distance l

In order to maintain reliable air-to-ground channels, the UAV in this scenario operates at a fixed altitude, while the horizontal distance between UAV and ground users is changed. The results clearly show that the feasible sum rate decreases with increasing user distance, mostly as a result of decreased signal strength and increased path loss. Nonetheless, a notable enhancement in performance is noted at all distances during the deployment of active RIS. In order to mitigate the extreme path loss that is commonly experienced in NLoS settings, the RIS intelligently reflects and focuses the UAV's broadcast signals toward the users, effectively increasing the received signal power. On the other hand, the sum rate rapidly diminishes in the UAV-based passive RIS system due to significant attenuation as the user gets farther away from the UAV. The RIS's ability to increase spectral efficiency and expand coverage in UAV-assisted communication networks is further supported by the fact that the performance difference between active RIS-assisted and passive RIS setups gets increasingly noticeable at greater distances. These results demonstrate how important RIS technology is for ensuring dependable connectivity during fixed-altitude UAV deployments, especially for users who are far away or on the edge [Fig. 4](#).

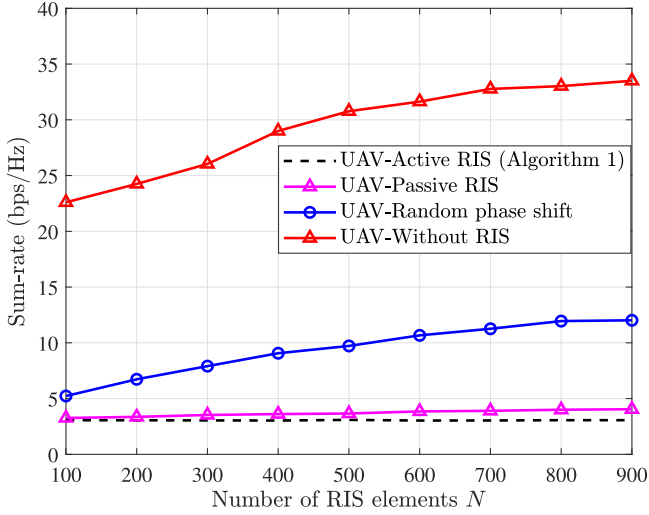


Fig. 3. Achieved sum rate versus the number of RIS elements.

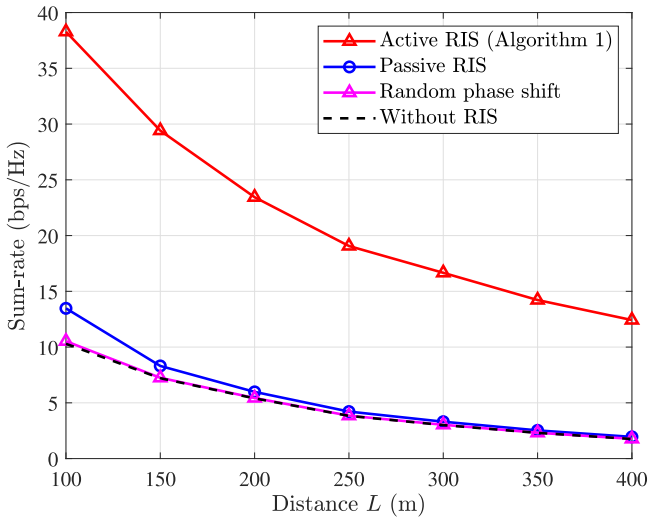


Fig. 4. Achieved sum rate versus the distance.

4.4. Sum rate performance with imperfect CSI

We examine the performance of a static UAV-aided active RIS with imperfect channel estimation. We define the imperfect channel estimates of hr , G , respectively:

$$\begin{aligned} \hat{hr} &= hr - \Delta hr \\ \hat{G} &= G - \Delta G \end{aligned} \quad (15)$$

where Δhr and ΔG describe the Channel Estimation Error (CEE) of hr and G , respectively. In the remainder of this study, we suppose that the CEEs pursue independent complex Gaussian distributions, i.e., $\Delta hr \simeq CN(0, \sigma_{\Delta hr}^2 I)$ and $\text{vec}(\Delta G) \simeq CN(0, \sigma_{\Delta G}^2 I)$. Additionally, the CEE variances are supposed to decline proportionally to the equivalent channel variances, which is:

$$\sigma_{\Delta hr_k}^2 = \frac{\epsilon}{N} \|hr_k\|^2, \sigma_{\Delta G}^2 = \frac{\epsilon}{N} \|G\|_F^2, \quad (16)$$

where $\epsilon \in [0, 1]$ characterizes the CEE level. Fig. 5 exhibits the sum rate of the static UAV-aided active RIS system according to the maximum transmit power $PT_{\max}$ within different error levels $\epsilon = 0, 0.1, 0.2, 0.4$.

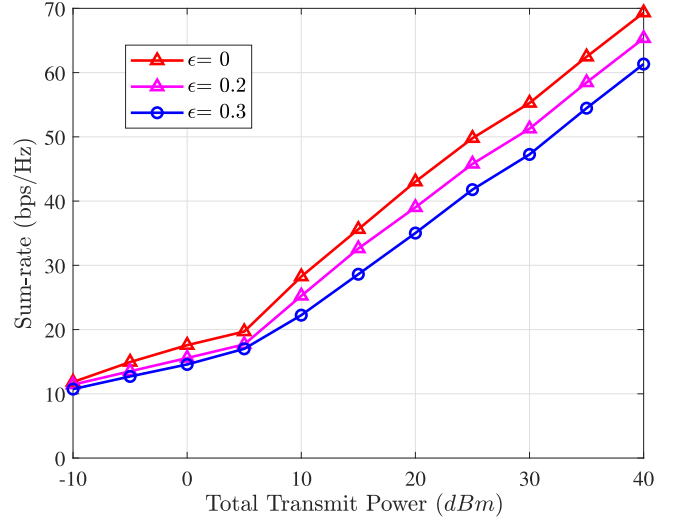


Fig. 5. Achieved sum rate versus the total power under imperfect channel state information with $\epsilon = 0, 0.2, 0.3$.

For different CEE levels, the sum rate increases as the transmit power rises since high power budgets guarantee an enhanced signal strength. Nevertheless, when ϵ increases, we notice that the sum rate diminishes, which is explained by the degradation generated by imperfect CSI. For example, when $PT_{\max} = 30 \text{ dBm}$, the sum rate is $SR = 55.26 \text{ bits/s/Hz}$ under a perfect channel while the $SR = 51.26$ under imperfect channel with $\epsilon = 0.2$. These results show that the suggested scheme is comparatively resilient to CEEs, retaining significant performance even in the face of suboptimal CSI settings.

4.5. Energy efficiency

We evaluate the energy efficiency performance of the proposed algorithm versus the total transmit power [34], defined as :

$$EE_{UAV}^{RIS} = \frac{SR}{P_{tot}^{RIS}}, EE_{UAV}^{AFR} = \frac{SR^{AFR}}{P_{tot}^{AFR}} \quad (17)$$

while, SR and SR^{AFR} represents the sum rate of the UAV-aided active RIS and AF relay-based systems, respectively. P_{tot}^{RIS} is the total power consumption of the UAV-aided active RIS system, described as:

$$P_{tot} = v_1 P_{UAV} + v_2 P_{RIS} + W_{UAV} + N W_S + N W_A + K W_k, \quad (18)$$

where v_1 and v_2 represent the energy conversion efficiencies at UAV and RIS, respectively. The consumed power by UAV transmitter, by each MS, by the phase-shift in RIS and the amplifier in active RIS elements are denoted respectively as: W_k, W_{UAV}, W_S, W_A . P_{tot}^{RIS} represents the total power consumption of the AF relay-aided system, which is written by:

$$P_{tot}^{AFR} = v_1 P_{UAV} + v_2 P_{AFR} + W_{UAV} + W_S^{AFR} + v_3 W_A^{AFR} + K W_k, \quad (19)$$

where P_{AFR} is the relay transmit power, W_A^{AFR} represents the power consumption per antenna element at the relay, v_3 is the number of AFR antennas W_S^{AFR} is the relay's static hardware power consumption.

Fig. 6 shows the maximum transmit power P as a function of the UAV EE. The UAV-active RIS-assisted system is found to consistently perform better than the UAV-passive RIS assisted system and AF relay, especially at low and medium P values. Additionally, for every scheme, the EE first rises with P before beginning to fall below a particular transmit power level. This unconventional variation can be explained as follows: at low transmit power levels, increasing P max significantly boosts the users' achievable rates, while the associated rise in power consumption remains relatively small. As a result, the energy efficiency improves. However, when P becomes sufficiently large, the system already exploits the full power budget to maximize the rate. In this high-power region,

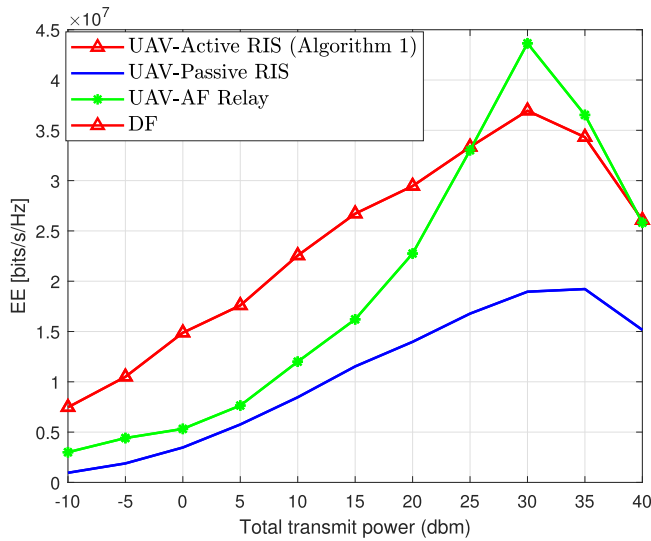


Fig. 6. Energy efficiency versus the total power.

further increases in lead to only marginal rate improvements, whereas the power consumption continues to rise, causing a sharp decline in energy efficiency.

Although active RIS has definite benefits for improving coverage and reducing path loss in UAV-assisted communication systems, its amplification capabilities adds extra power consumption that has a direct impact on overall efficiency. Active RIS depends on incorporated amplifiers and control circuitry, in contrast to passive RIS, which only adjusts incident signals excluding requiring for external power. The system's overall power budget rises as a result of these components' constant energy use. The advantages of active RIS must be balanced against this overhead from an efficiency aspect. The amplification increases the power of the received signal, lowers the risk of an outage, and increases coverage, all of which can result in fewer retransmissions and less communication energy. However, in UAV settings where energy resources are limited by battery capacity and flight endurance, the additional demand from active elements may counteract these gains. Since propulsion energy already dominates the system's power profile, this trade-off becomes more noticeable when UAV mobility is taken into account. As a result, including active RIS into UAV frameworks reveals two challenges: improving performance while preserving energy efficiency. To balance signal augmentation with sustainable energy consumption, future research should investigate energy-aware amplifier designs, adaptive activation of RIS elements, and cooperative optimization of UAV trajectory and RIS operation. These strategies will be essential to maintaining the effectiveness and practicality of active RIS-assisted UAV systems in actual deployments.

5. Conclusion

In this study, we examined a downlink communication system in which an active RIS supports a static UAV functioning as a base station. We evaluated the system's sum rate, energy efficiency, and tolerance to channel estimate errors. The results obtained show that, as compared to a passive RIS, the implementation of an active RIS greatly increases the achievable total rate because it can compensate for substantial propagation losses commonly found in UAV-assisted links and provide additional signal amplification. Moreover, despite the greater power consumption imposed by the amplification, the active RIS system achieves higher energy efficiency in the low-to-moderate transmit power regime. The performance under channel estimate errors reveals that the active RIS architecture is more robust and can maintain accurate communication even in imperfect CSI situations. Overall, the results show that

active RIS is an exciting prospect for improving UAV-assisted wireless networks, especially in energy-constrained or coverage-limited scenarios, whereas passive RIS remains an appealing option when reducing hardware energy consumption is the primary goal. A more thorough power consumption model that simultaneously takes into consideration RIS hardware features and UAV flight dynamics is a crucial area for further study. Such an extension would assist close the gap between theoretical performance and real-world deployment and offer a more accurate evaluation of energy efficiency in UAV-RIS systems.

CRedit authorship contribution statement

Wafa Abid: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization; **Moufida Hajjaj:** Writing – review & editing, Conceptualization; **Ridha Bouallegue:** Supervision.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no conflicts of interest and there are no funding sources or grant IDs to acknowledge for this research.

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